Signals and Systems
Assignment-2: Solution

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Exercise 1 (1)

Calculate: $\int_{-\infty}^{+\infty} \delta(t-5)\cos(\pi t)\mathrm{d}t$

$$\begin{aligned}\int_{-\infty}^{+\infty} \delta(t-5)\cos(\pi t)\mathrm{d}t &= \int_{-\infty}^{+\infty} \delta(t-5)\cos(5\pi)\mathrm{d}t \\ &= \cos(5\pi)\underbrace{\int_{-\infty}^{+\infty} \delta(t-5)\mathrm{d}t}_{=1 \text{ since } 5\in[-\infty \quad \infty]} \\ &= \cos(5\pi) = -1\end{aligned}$$

Calculate: $\int_0^1 \delta(-2t+1)\mathrm{d}t$

$$\begin{aligned}\int_0^1 \delta(-2t+1)\mathrm{d}t &= \int_0^1 \delta\left(-2(t-\frac{1}{2})\right)\mathrm{d}t \\ &= \frac{1}{2}\underbrace{\int_0^1 \delta\left(t-\frac{1}{2}\right)\mathrm{d}t}_{=1 \text{ since } 1/2\in[0 \quad 1]} = \frac{1}{2}\end{aligned}$$

Exercises 1 (2)

Calculate: $\int_0^{20} \delta(t-8)\mathrm{rect}(\frac{t}{16})\mathrm{d}t$

$$\int_0^{20} \delta(t-8)\mathrm{rect}\left(\frac{t}{16}\right)\mathrm{d}t = \int_0^{20} \delta(t-8)\underbrace{\mathrm{rect}\left(\frac{8}{16}\right)}_{=1}\mathrm{d}t$$

$$= \int_0^{20} \delta(t-8)\mathrm{d}t = 1 \;\text{ since }\; 8 \in [0 \;\; 20]$$

Calculate: $\int_{20}^{40} \delta(t-8)\mathrm{rect}(\frac{t}{16})\mathrm{d}t$

$$\int_{20}^{40} \delta(t-8)\mathrm{rect}\left(\frac{t}{16}\right)\mathrm{d}t = \int_{20}^{40} \delta(t-8)\underbrace{\mathrm{rect}\left(\frac{8}{16}\right)}_{=1}\mathrm{d}t$$

$$= \int_{20}^{40} \delta(t-8)\mathrm{d}t = 0 \;\text{ since }\; 8 \notin [20 \;\; 40]$$

Exercise 1 (3)

Calculate: $\int_0^{20} \delta(t-8)\mathrm{tri}(\frac{t}{32})\mathrm{d}t$

$$\int_0^{20} \delta(t-8)\mathrm{tri}\left(\frac{t}{32}\right)\mathrm{d}t = \int_0^{20} \delta(t-8)\mathrm{tri}\left(\frac{8}{32}\right)\mathrm{d}t$$

$$= \mathrm{tri}\left(\frac{8}{32}\right) \underbrace{\int_0^{20} \delta(t-8)\mathrm{d}t}_{=1 \quad \text{since} \quad 8\in[0 \quad 20]}$$

$$= \mathrm{tri}\left(\frac{8}{32}\right) = \mathrm{tri}\left(\frac{1}{4}\right)$$

Given that $0 \leq \frac{1}{4} \leq 1$ and $\mathrm{tri}(t) = 1 - t$ for $0 \leq t \leq 1$, then:

$$\int_0^{20} \delta(t-8)\mathrm{tri}\left(\frac{t}{32}\right)\mathrm{d}t = \mathrm{tri}\left(\frac{1}{4}\right) = 1 - \frac{1}{4} = \frac{3}{4}$$

Exercise 1 (4)

Calculate: $\int_{-\infty}^{+\infty} \mathrm{sinc}(t)\mathrm{comb}(t)\mathrm{d}t$

$$\int_{-\infty}^{+\infty} \mathrm{sinc}(t)\mathrm{comb}(t)\mathrm{d}t = \int_{-\infty}^{+\infty} \mathrm{sinc}(t) \left[\sum_{n\in\mathbb{Z}} \delta(t-n)\right] \mathrm{d}t$$

$$= \int_{-\infty}^{+\infty} \left[\sum_{n\in\mathbb{Z}} \mathrm{sinc}(t)\delta(t-n)\right] \mathrm{d}t$$

$$= \int_{-\infty}^{+\infty} \left[\sum_{n\in\mathbb{Z}} \mathrm{sinc}(n)\delta(t-n)\right] \mathrm{d}t$$

$$= \int_{-\infty}^{+\infty} \left[\underbrace{\mathrm{sinc}(0)}_{=1}\delta(t) + \sum_{n\in\mathbb{Z}\ ,\ n\neq 0} \underbrace{\mathrm{sinc}(n)}_{=0 \text{ for } n\in\mathbb{Z}^*} \delta(t-n)\right] \mathrm{d}t$$

$$= \int_{-\infty}^{+\infty} \delta(t)\mathrm{d}t = 1$$

Exercise 1 (5)

Calculate: $I = \int_{-\infty}^{+\infty} \mathrm{rect}\left(\frac{t}{4}\right) \mathrm{comb}\left(\frac{2t}{3}\right) \mathrm{d}t$

- Using the relation $\mathrm{comb}(at) = \frac{1}{|a|} \sum_n \delta(t - n/a)$:

$$
\begin{aligned}
I &= \int_{-\infty}^{+\infty} \mathrm{rect}\left(\frac{t}{4}\right) \left[\frac{3}{2} \sum_{n\in\mathbb{Z}} \delta\left(t - \frac{3}{2}n\right)\right] \mathrm{d}t \\
&= \frac{3}{2} \int_{-\infty}^{+\infty} \sum_{n\in\mathbb{Z}} \left[\mathrm{rect}\left(\frac{t}{4}\right) \delta\left(t - \frac{3}{2}n\right)\right] \mathrm{d}t \\
&= \frac{3}{2} \int_{-\infty}^{+\infty} \sum_{n\in\mathbb{Z}} \left[\mathrm{rect}\left(\frac{3n}{8}\right) \delta\left(t - \frac{3}{2}n\right)\right] \mathrm{d}t
\end{aligned}
$$

- $\mathrm{rect}\left(\frac{3n}{8}\right)$ is different from zero (and equal to 1) for:

$$
-\frac{1}{2} \leq \frac{3n}{8} \leq \frac{1}{2} \;\Rightarrow\; -\frac{4}{3} \leq n \leq \frac{4}{3} \;\Rightarrow\; n = 0, \pm 1
$$

- Consequently:

$$
I = \frac{3}{2} \int_{-\infty}^{+\infty} \left[\delta\left(t + \frac{3}{2}\right) + \delta(t) + \delta\left(t - \frac{3}{2}\right)\right] \mathrm{d}t = \frac{3}{2} \times 3 = \frac{9}{2}
$$

Exercise 1 (6)

Calculate: $I = \int_{-\infty}^{+\infty} \mathrm{tri}\left(\frac{t}{2}\right) \mathrm{comb}\,(t)\,\mathrm{d}t$

- The above integral can be written as:

$$I = \int_{-\infty}^{+\infty} \mathrm{tri}\left(\frac{t}{2}\right) \sum_{n\in\mathbb{Z}} \delta(t-n)\mathrm{d}t$$

$$= \int_{-\infty}^{+\infty} \left[\sum_{n\in\mathbb{Z}} \mathrm{tri}\left(\frac{n}{2}\right) \delta(t-n)\right] \mathrm{d}t$$

- $\mathrm{tri}\left(\frac{n}{2}\right)$ is different from zero for:

$$-1 < \frac{n}{2} < 1 \;\Rightarrow\; -2 < n < 2 \;\Rightarrow\; n = 0, \pm 1$$

- Consequently:

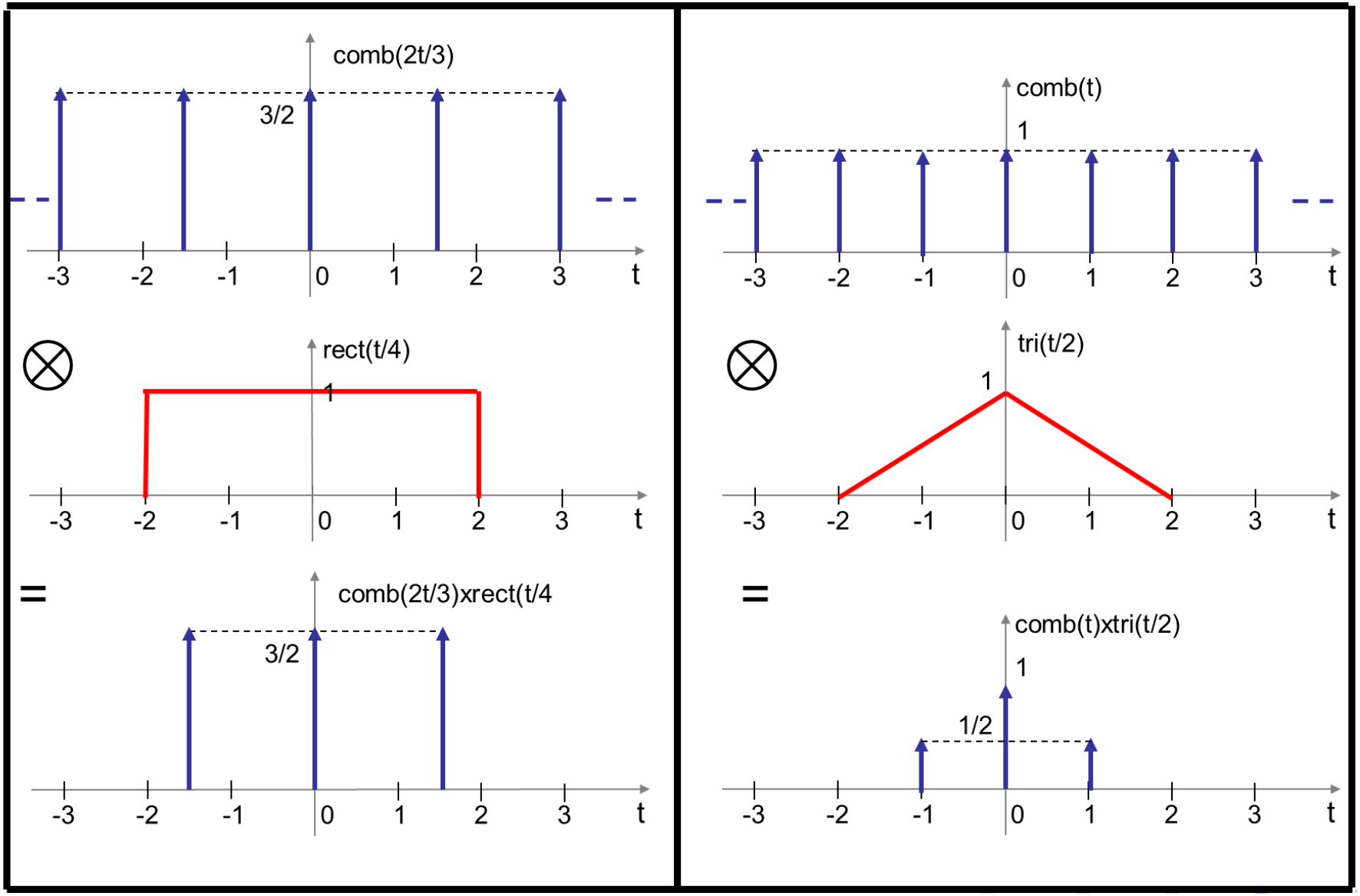
$$I = \int_{-\infty}^{+\infty} \left[\mathrm{tri}\left(-\frac{1}{2}\right) \delta(t+1) + \mathrm{tri}(0)\delta(t) + \mathrm{tri}\left(\frac{1}{2}\right) \delta(t-1)\right] \mathrm{d}t$$

$$= \frac{1}{2} + 1 + \frac{1}{2} = 2$$

where the relation $\mathrm{tri}(t) = 1 - |t|$ for $-1 \leq t \leq 1$ was used.

Exercise 1 (7)

- The products in the last two parts are better illustrated in the figure below:



Exercise 1 (8)

Calculate: $I = \int_{0.5}^{2.5} \mathrm{rect}\left(\frac{t}{3}\right) \mathrm{comb}\,(t)\,\mathrm{d}t$

<u>Method 1:</u>

- The above integral can be written as:

$$I = \int_{0.5}^{2.5} \sum_{n=-\infty}^{+\infty} \mathrm{rect}\left(\frac{n}{3}\right) \delta\,(t-n)\,\mathrm{d}t$$

  where $\mathrm{rect}\left(\frac{n}{3}\right) = 1$ for $-\frac{1}{2} \leq \frac{n}{3} \leq \frac{1}{2}$ implying that $-\frac{3}{2} \leq n \leq \frac{3}{2}$ implying that $n = -1, 0, 1$.

- Consequently:

$$\begin{aligned} I &= \int_{0.5}^{2.5} [\delta(t+1) + \delta(t) + \delta(t-1)]\,\mathrm{d}t \\ &= \underbrace{\int_{0.5}^{2.5} \delta(t+1)\mathrm{d}t}_{=0\ ;\ -1\notin[0.5\ 2.5]} + \underbrace{\int_{0.5}^{2.5} \delta(t)\mathrm{d}t}_{=0\ ;\ 0\notin[0.5\ 2.5]} + \underbrace{\int_{0.5}^{2.5} \delta(t-1)\mathrm{d}t}_{=1\ ;\ 1\in[0.5\ 2.5]} \\ &= 0 + 0 + 1 = 1 \end{aligned}$$

Exercise 1 (9)

Calculate: $I = \int_{0.5}^{2.5} \mathrm{rect}\left(\frac{t}{3}\right) \mathrm{comb}\left(t\right) \mathrm{d}t$

<u>Method 2:</u>

- The above integral can be written as:

$$I = \int_{0.5}^{2.5} \sum_{n=-\infty}^{+\infty} \mathrm{rect}\left(\frac{n}{3}\right) \delta\left(t-n\right) \mathrm{d}t$$

$$= \sum_{n=-\infty}^{+\infty} \mathrm{rect}\left(\frac{n}{3}\right) \int_{0.5}^{2.5} \delta\left(t-n\right) \mathrm{d}t$$

where $\int_{0.5}^{2.5} \delta\left(t-n\right) \mathrm{d}t = 1$ for $n \in [0.5 \;\; 2.5]$ $(n = 1, 2)$ and equal to zero elsewhere.

- Consequently:

$$I = \underbrace{\mathrm{rect}\left(\frac{1}{3}\right)}_{=1\ ;\ \frac{1}{3}\in\left[-\frac{1}{2}\ \ \frac{1}{2}\right]} + \underbrace{\mathrm{rect}\left(\frac{2}{3}\right)}_{=0\ ;\ \frac{2}{3}\notin\left[-\frac{1}{2}\ \ \frac{1}{2}\right]}$$

$$= 1 + 0 = 1$$

Exercise 2 (a)

- In order to take advantage of the relation: $\frac{1}{a}\text{rect}\left(\frac{t}{a}\right) * \frac{1}{a}\text{rect}\left(\frac{t}{a}\right) = \frac{1}{a}\text{tri}\left(\frac{t}{a}\right)$, the signal $x(t)$ can be written as:

$$
\begin{aligned}
x(t) &= 4\left[\frac{1}{2}\text{rect}\left(\frac{t}{2}\right) * \delta(t+1)\right] * \frac{1}{2}\text{rect}\left(\frac{t}{2}\right) \\
&= 4\left[\frac{1}{2}\text{rect}\left(\frac{t}{2}\right) * \frac{1}{2}\text{rect}\left(\frac{t}{2}\right)\right] * \delta(t+1) \\
&= 4\left[\frac{1}{2}\text{tri}\left(\frac{t}{2}\right)\right] * \delta(t+1) \\
&= 2\text{tri}\left(\frac{t+1}{2}\right)
\end{aligned}
$$

Exercise 2 (b)

- $x(t)$ can be written as:

$$
\begin{aligned}
x(t) &= 4\left[\frac{1}{2}\mathrm{rect}\left(\frac{t}{2}\right) * \delta(t-1)\right] * \left[\frac{1}{2}\mathrm{rect}\left(\frac{t}{2}\right) * \delta(t+3)\right] \\
&= 4\left[\frac{1}{2}\mathrm{rect}\left(\frac{t}{2}\right) * \frac{1}{2}\mathrm{rect}\left(\frac{t}{2}\right)\right] * [\delta(t-1) * \delta(t+3)] \\
&= 4\left[\frac{1}{2}\mathrm{tri}\left(\frac{t}{2}\right)\right] * \delta((t+3)-1) \\
&= 2\mathrm{tri}\left(\frac{t}{2}\right) * \delta(t+2) \\
&= 2\mathrm{tri}\left(\frac{t+2}{2}\right)
\end{aligned}
$$

Exercise 3 (1)

$y(t) = \int_{-\infty}^{t} x(\tau)\mathrm{d}\tau$

- The impulse response is the response of the system when the input is equal to the unit impulse. Replacing $x(\tau)$ by $\delta(\tau)$ we obtain:

$$h(t) = \int_{-\infty}^{t} \delta(\tau)\mathrm{d}\tau = \int_{-\infty}^{t} \delta(\tau - 0)\mathrm{d}\tau$$

  - if $0 \in [-\infty \;\; t]$ (in other words $t > 0$), then $h(t) = 1$.
  - if $0 \notin [-\infty \;\; t]$ (in other words $t \leq 0$), then $h(t) = 0$.

Therefore: $h(t)$ is equal to the unit step function:

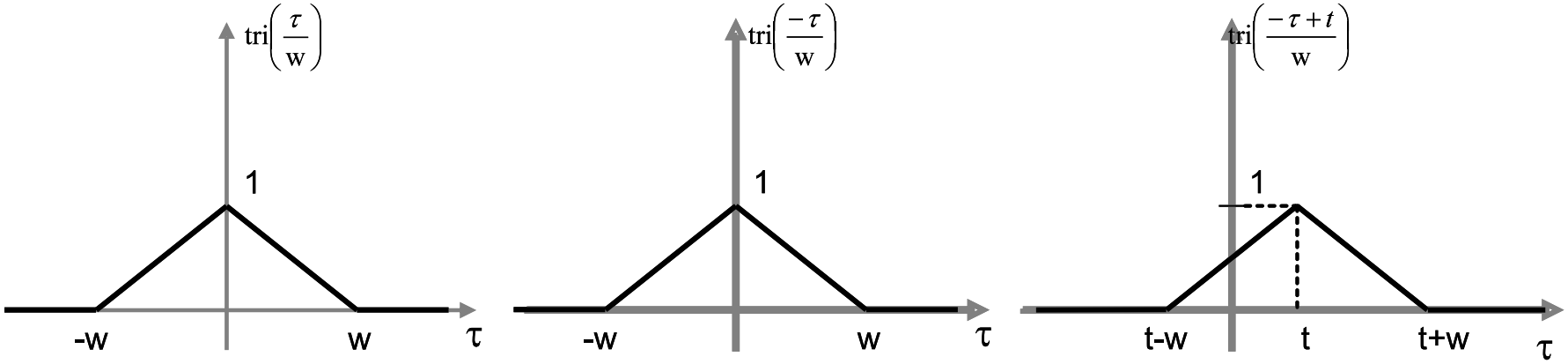
$$h(t) = u(t)$$

Exercise 3 (2)

Determine the output when: $x(t) = \mathrm{tri}\left(\frac{t}{w}\right)$.

- The output can be determined from the convolution of the input $x(t)$ with the impulse response $h(t)$. In other words:

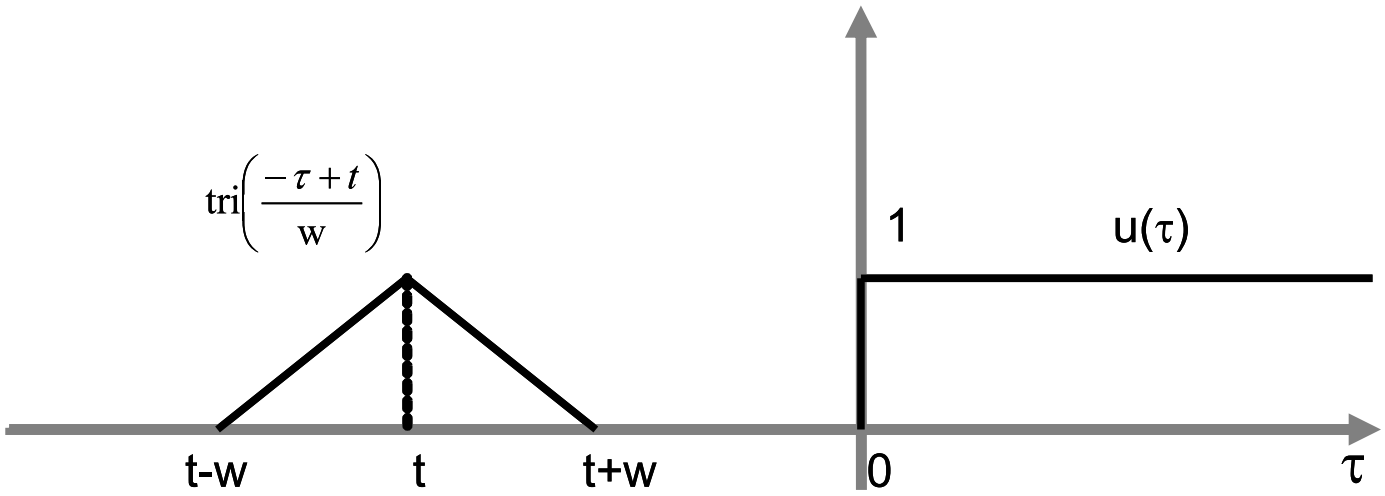
$$y(t) = h(t) * x(t) = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)\mathrm{d}\tau$$



Exercise 3 (3)

Case 1:

$$t + w < 0 \;\Rightarrow\; t < -w$$



In this case:

$$\mathrm{tri}\left(\frac{t-\tau}{w}\right) u(\tau) = 0$$

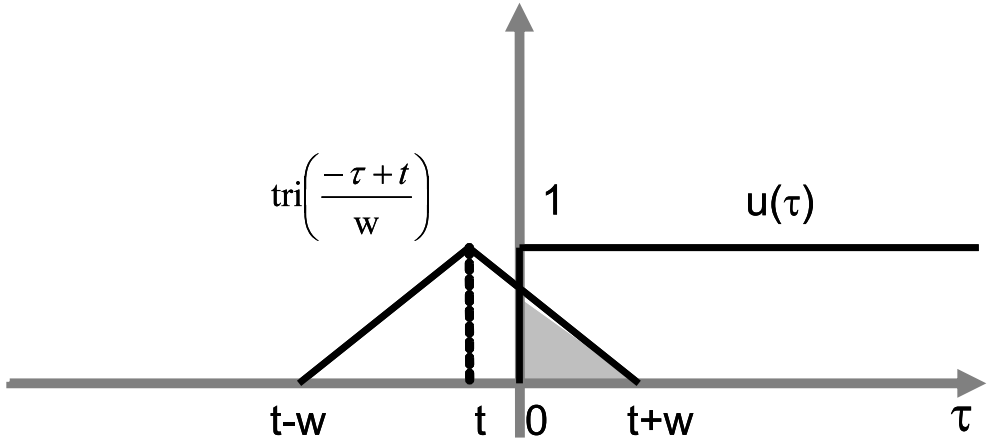
and:

$$y(t) = 0$$

Exercise 3 (4)

Case 2:

$$t \leq 0 \leq t + w \;\Rightarrow\; -w \leq t \leq 0$$



In this case:

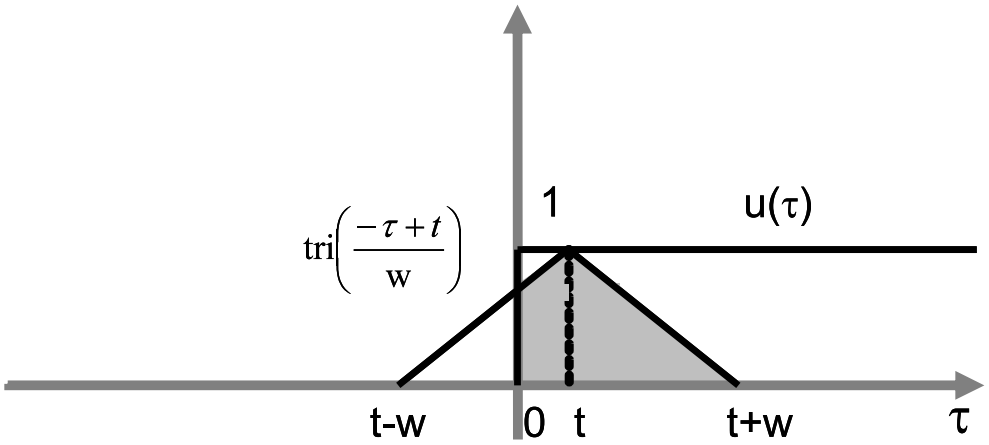
$$y(t) = \int_0^{t+w} \left(\frac{-\tau + t}{w} + 1 \right) \mathrm{d}\tau$$

$$= \left[-\frac{\tau^2}{2w} + \left(\frac{t+w}{w} \right) \tau \right]_0^{t+w} = \cdots = \frac{(t+w)^2}{2w}$$

Exercise 3 (5)

Case 3:

$$t - w \leq 0 \leq t \;\Rightarrow\; 0 \leq t \leq w$$



In this case:

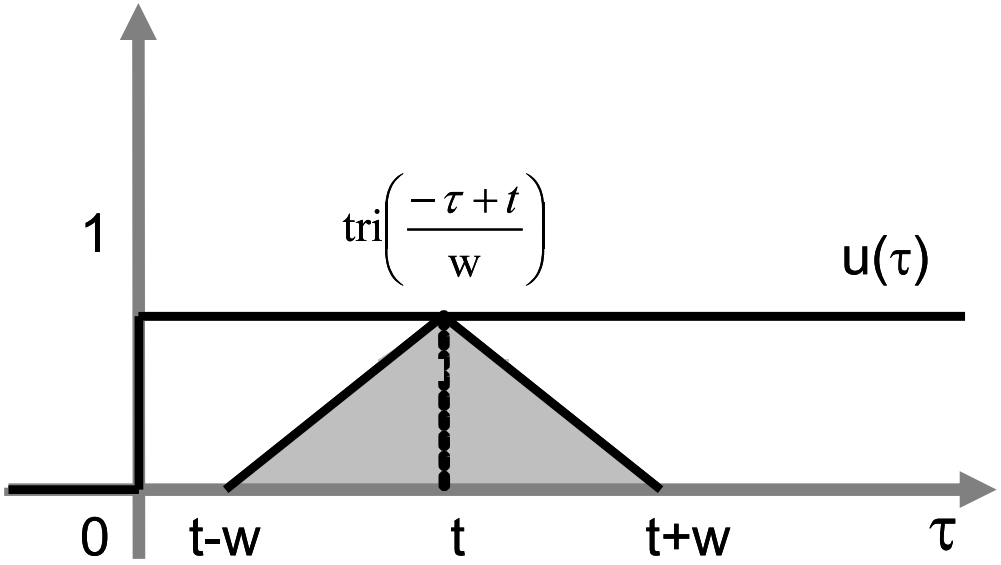
$$y(t) = \int_0^t \left(\frac{\tau - t}{w} + 1\right) \mathrm{d}\tau + \underbrace{\int_t^{t+w} \left(\frac{-\tau + t}{w} + 1\right) \mathrm{d}\tau}_{=\frac{w}{2}}$$

$$= \left[\frac{\tau^2}{2w} + \left(\frac{-t + w}{w}\right)\tau\right]_0^t + \frac{w}{2} = \cdots = \frac{-t^2 + 2wt + w^2}{2w}$$

Exercise 3 (6)

Case 4:

$$0 \leq t - w \;\Rightarrow\; t \geq w$$



In this case:

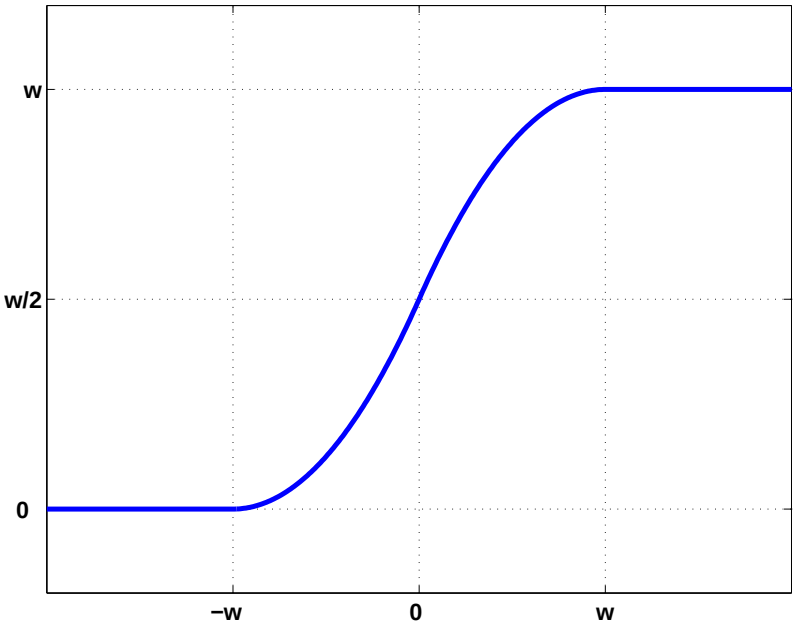
$$y(t) = \underbrace{\int_{t-w}^{t} \left(\frac{\tau - t}{w} + 1\right) \mathrm{d}\tau}_{=\frac{w}{2}} + \underbrace{\int_{t}^{t+w} \left(\frac{-\tau + t}{w} + 1\right) \mathrm{d}\tau}_{=\frac{w}{2}}$$

$$= w$$

Exercise 3 (7)

- Finally, $y_(t)$ is given by:

$$y(t) = \begin{cases} y_1(t) = 0, & t \leq -w; \\ y_2(t) = \frac{(t+w)^2}{2w}, & -w \leq t \leq 0; \\ y_3(t) = \frac{-t^2+2wt+w^2}{2w}, & 0 \leq t \leq w;; \\ y_4(t) = w, & t \geq w. \end{cases}$$



- Note that the boundary conditions are satisfied:

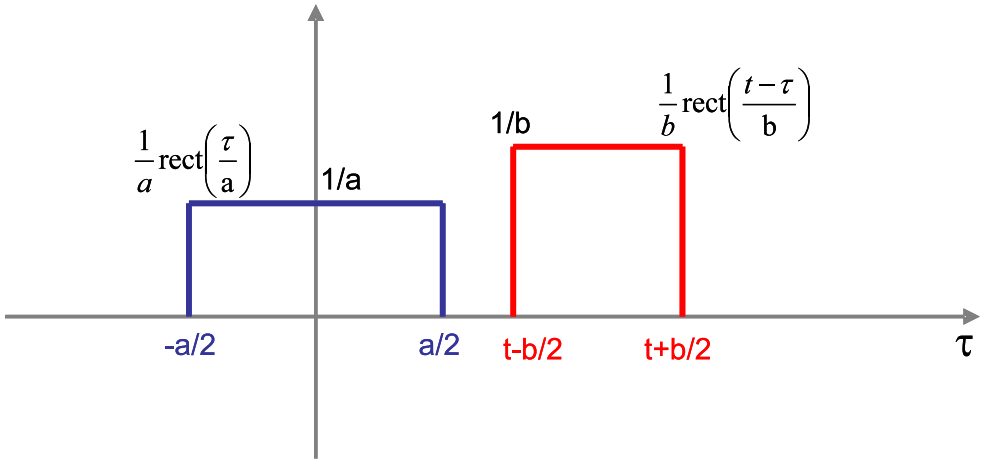
$$\begin{aligned} y_1(-w) &= y_2(-w) = 0 \\ y_2(0) &= y_3(0) = \frac{w}{2} \\ y_3(w) &= y_4(w) = w \end{aligned}$$

Exercise 4 (1)

- $y(t)$ is given by:

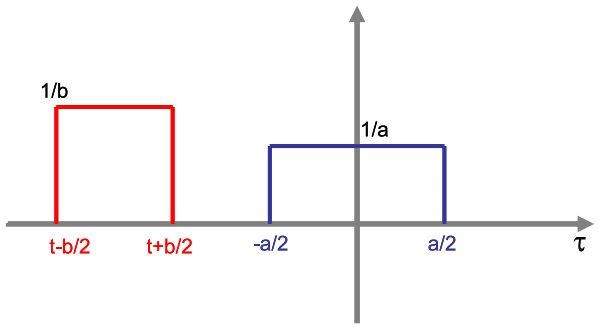
$$y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{+\infty} \frac{1}{a}\mathrm{rect}\left(\frac{\tau}{a}\right) . \frac{1}{b}\mathrm{rect}\left(\frac{t-\tau}{b}\right) \mathrm{d}\tau$$

- The concerned signals are shown in the figure below:



Exercise 4 (2)

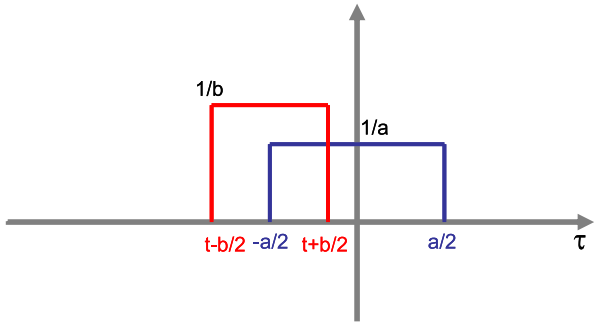
- Case 1:



  - $t + \frac{b}{2} \leq -\frac{a}{2}$ implying that $t \leq -\frac{a}{2} - \frac{b}{2}$:

$$y(t) = 0$$

- Case 2:

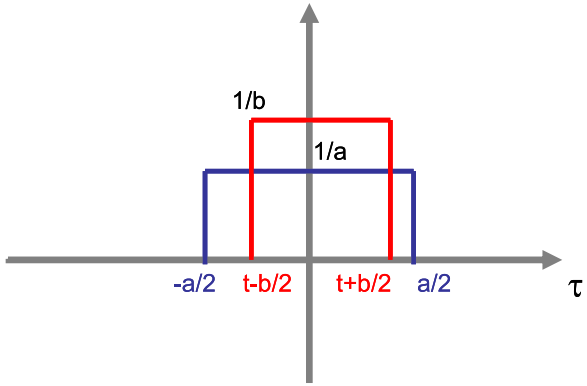


  - $t - \frac{b}{2} \leq -\frac{a}{2} \leq t + \frac{b}{2}$ implying that $-\frac{a}{2} - \frac{b}{2} \leq t \leq -\frac{a}{2} + \frac{b}{2}$:

$$y(t) = \frac{1}{ab}\left(t + \frac{a}{2} + \frac{b}{2}\right)$$

Exercise 4 (3)

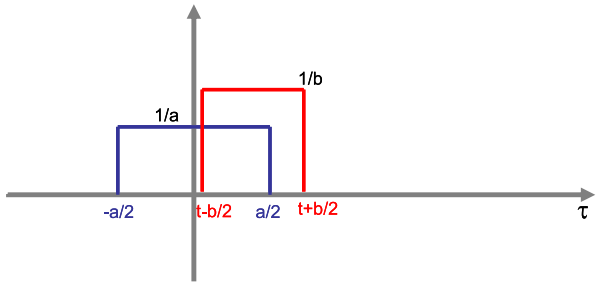
- Case 3:



  - $t - \frac{b}{2} \geq -\frac{a}{2}$ and $t + \frac{b}{2} \leq \frac{a}{2}$ implying that $-\frac{a}{2} + \frac{b}{2} \leq t \leq \frac{a}{2} - \frac{b}{2}$:

$$y(t) = \frac{1}{ab} b = \frac{1}{a}$$

- Case 4:

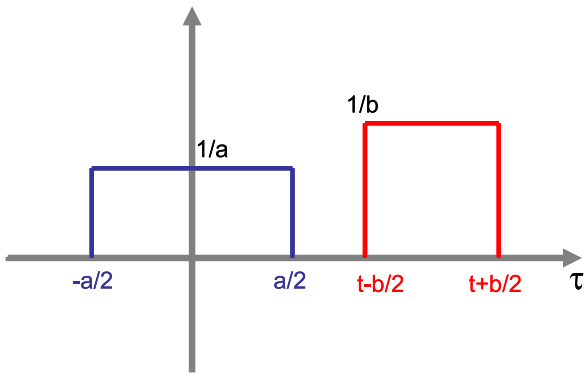


  - $t - \frac{b}{2} \leq \frac{a}{2} \leq t + \frac{b}{2}$ implying that $\frac{a}{2} - \frac{b}{2} \leq t \leq \frac{a}{2} + \frac{b}{2}$:

$$y(t) = \frac{1}{ab}\left(\frac{a}{2} - \left(t - \frac{b}{2}\right)\right) = \frac{1}{ab}\left(-t + \frac{a}{2} + \frac{b}{2}\right)$$

Exercise 4 (4)

- Case 5:



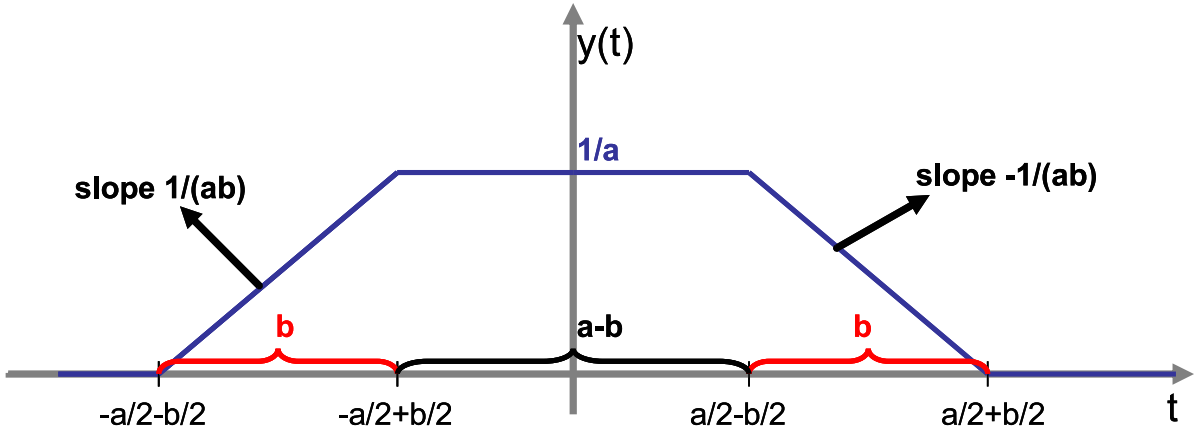
  - $t - \frac{b}{2} \geq \frac{a}{2}$ implying that $t \geq \frac{a}{2} + \frac{b}{2}$:

$$y(t) = 0$$

Exercise 4 (5)

- Finally, $y(t)$ is given by:

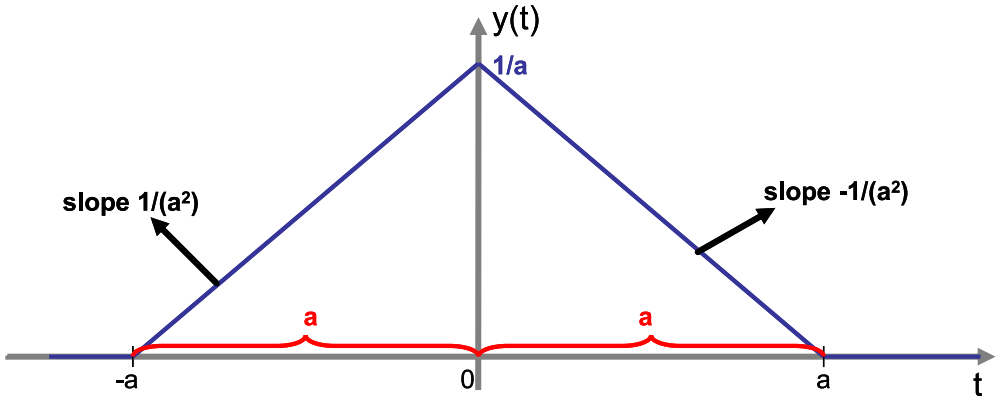
$$y(t) = \begin{cases} 0, & t \leq -\frac{a}{2} - \frac{b}{2}; \\ \frac{1}{ab}\left(t + \frac{a}{2} + \frac{b}{2}\right), & -\frac{a}{2} - \frac{b}{2} \leq t \leq -\frac{a}{2} + \frac{b}{2}; \\ \frac{1}{a}, & -\frac{a}{2} + \frac{b}{2} \leq t \leq \frac{a}{2} - \frac{b}{2}; \\ \frac{1}{ab}\left(-t + \frac{a}{2} + \frac{b}{2}\right), & \frac{a}{2} - \frac{b}{2} \leq t \leq \frac{a}{2} + \frac{b}{2}; \\ 0, & t \geq \frac{a}{2} + \frac{b}{2}. \end{cases}$$



Exercise 4 (6)

- For $a = b$, $y(t)$ reduces to:

$$y(t) = \begin{cases} 0, & t \leq -a; \\ \frac{1}{a^2}(t+a), & -a \leq t \leq 0; \\ \frac{1}{a}, & 0 \leq t \leq 0; \\ \frac{1}{a^2}(-t+a), & 0 \leq t \leq a; \\ 0, & t \geq a. \end{cases}$$



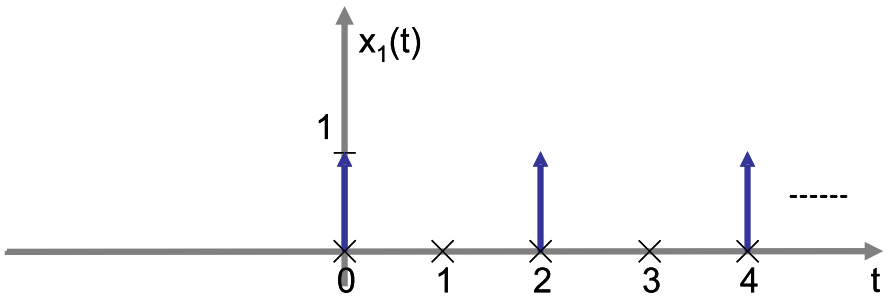
- This is in agreement with the result obtained in the course:

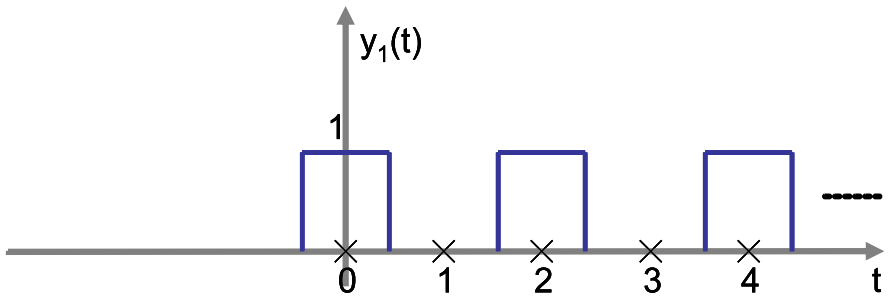
$$\frac{1}{a}\mathrm{rect}\left(\frac{t}{a}\right) * \frac{1}{a}\mathrm{rect}\left(\frac{t}{a}\right) = \frac{1}{a}\mathrm{tri}\left(\frac{t}{a}\right)$$

Exercise 5 (1)

- On the upper branch, $x_1(t) = \sum_{n\in\mathbb{N}} \delta(t-2n)$ results in the following output:

$$y_1(t) = \mathrm{rect}(t) * x_1(t) = \mathrm{rect}(t) * \sum_{n\in\mathbb{N}} \delta(t-2n)$$
$$= \sum_{n\in\mathbb{N}} \mathrm{rect}(t) * \delta(t-2n) = \sum_{n\in\mathbb{N}} \mathrm{rect}(t-2n)$$

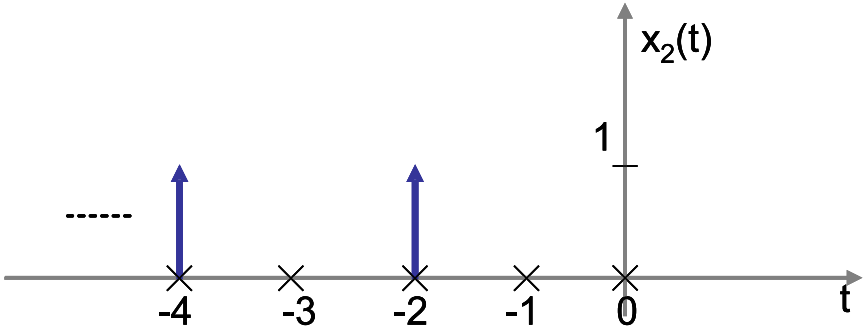


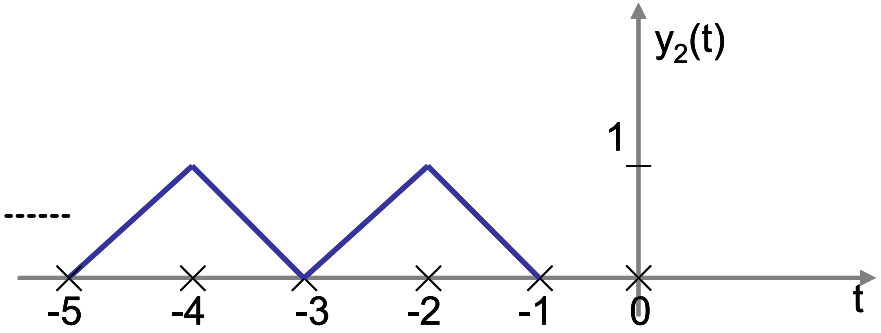


Exercise 5 (2)

- On the lower branch, the impulse response of the concatenation of the blocks is $\mathrm{rect}(t) * \mathrm{rect}(t) = \mathrm{tri}(t)$.
- Consequently, the input $x_2(t) = \sum_{n\in\mathbb{N},n\neq 0} \delta(t+2n)$ results in the following output:

$$y_2(t) = \mathrm{tri}(t) * x_2(t) = \mathrm{tri}(t) * \sum_{n\in\mathbb{N},n\neq 0} \delta(t+2n)$$
$$= \sum_{n\in\mathbb{N},n\neq 0} \mathrm{tri}(t) * \delta(t+2n) = \sum_{n\in\mathbb{N},n\neq 0} \mathrm{tri}(t+2n)$$

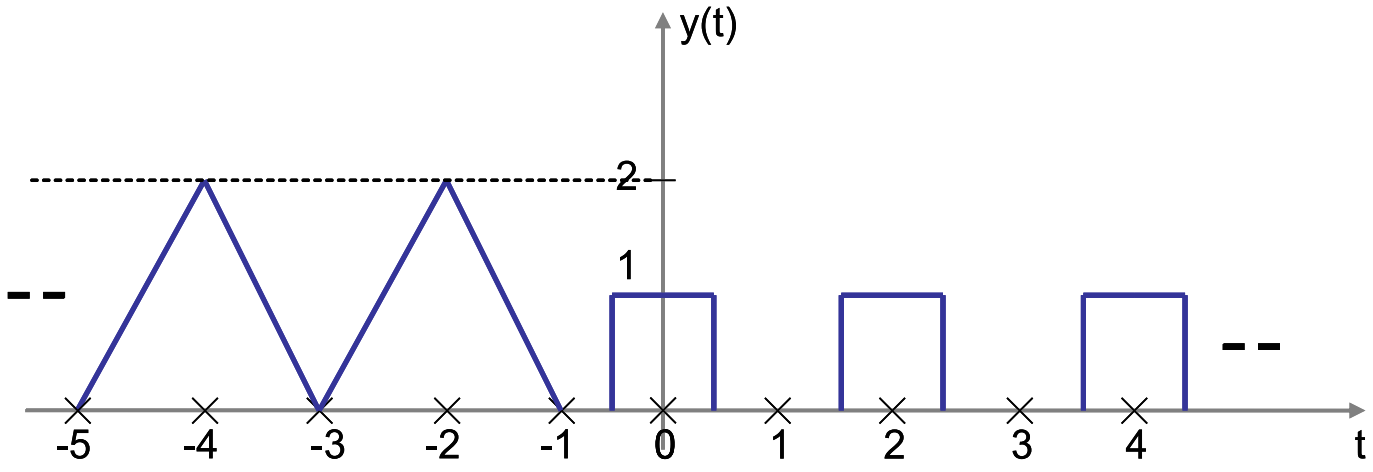




Exercise 5 (3)

- Finally, the output $y(t)$ is given by:

$$
\begin{aligned}
y(t) &= y_1(t) + 2y_2(t) \\
&= \sum_{n\in\mathbb{N}} \mathrm{rect}(t-2n) + 2\sum_{n\in\mathbb{N}} \mathrm{tri}(t+2n)
\end{aligned}
$$

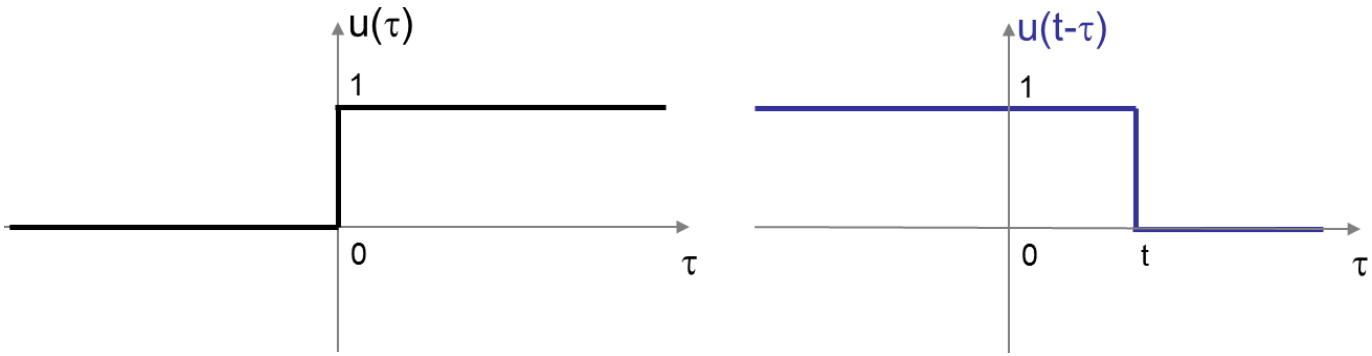


Exercise 6 (1)

- The step response $y(t)$ can be written as:

$$y(t) = u(t) * h(t) = 2u(t) * u(t) - u(t) * e^{-(t-2)}u(t-2)$$

- First we calculate $y_1(t) = u(t) * u(t) = \int_{-\infty}^{+\infty} u(t-\tau)u(\tau)\mathrm{d}\tau$:



  - Case 1: $t \leq 0$, $y_1(t) = 0$.
  - Case 2: $t \geq 0$:

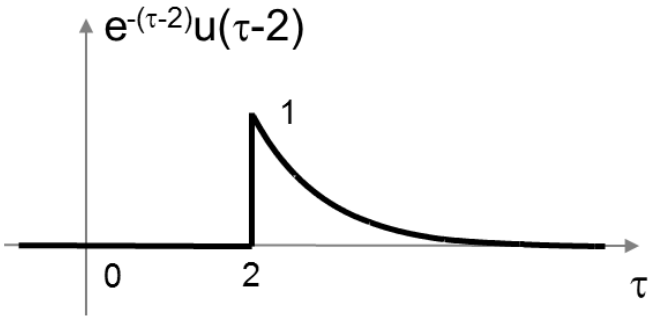
$$y_1(t) = \int_0^t (1 \times 1)\mathrm{d}\tau = t$$

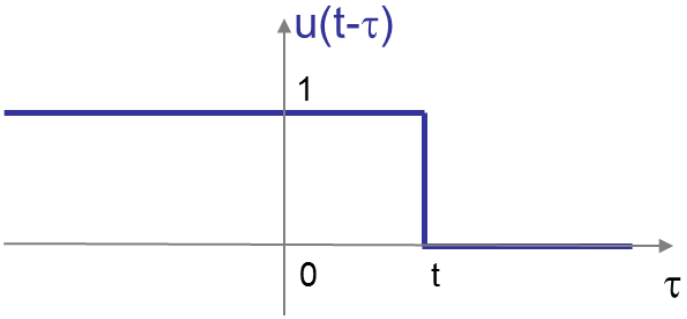
  - Finally:

$$y_1(t) = \begin{cases} 0, & t \leq 0; \\ t, & t \geq 0. \end{cases} = tu(t) = \mathrm{ramp}(t)$$

Exercise 6 (2)

- At a second time, we calculate $y_2(t) = u(t) * e^{-(t-2)}u(t-2) = \int_{-\infty}^{+\infty} u(t-\tau)e^{-(\tau-2)}u(\tau-2)\mathrm{d}\tau$:





  - Case 1: $t \leq 2$, $y_2(t) = 0$.
  - Case 2: $t \geq 2$:

$$y_2(t) = \int_2^t (1 \times e^{-(\tau-2)})\mathrm{d}\tau = \left[-e^{-(\tau-2)}\right]_2^t = 1 - e^{-(t-2)}$$

  - Finally:

$$y_2(t) = \begin{cases} 0, & t \leq 2; \\ 1 - e^{-(t-2)}, & t \geq 2. \end{cases} = \left[1 - e^{-(t-2)}\right] u(t-2)$$

Exercise 6 (3)

- Finally:

$$
\begin{aligned}
y(t) &= 2y_1(t) - y_2(t) \\
&= 2tu(t) + \left[e^{-(t-2)} - 1\right] u(t-2)
\end{aligned}
$$

Exercise 6 (4)

- $x(t) = \mathrm{rect}(2t - 2)$ can be written as:

$$x(t) = u\left(t - \frac{3}{4}\right) - u\left(t - \frac{5}{4}\right)$$

- Since the system is linear, then the response to $x(t)$ can be written as:

$$y_1(t) = y\left(t - \frac{3}{4}\right) - y\left(t - \frac{5}{4}\right)$$

where $y(t) = 2tu(t) + [e^{-(t-2)} - 1]u(t-2)$ is the step response.

- Consequently:

$$y_1(t) = \left[2\left(t - \frac{3}{4}\right)u\left(t - \frac{3}{4}\right) + [e^{-(t-11/4)} - 1]u\left(t - \frac{11}{4}\right)\right]$$
$$- \left[2\left(t - \frac{5}{4}\right)u\left(t - \frac{5}{4}\right) + [e^{-(t-13/4)} - 1]u\left(t - \frac{13}{4}\right)\right]$$